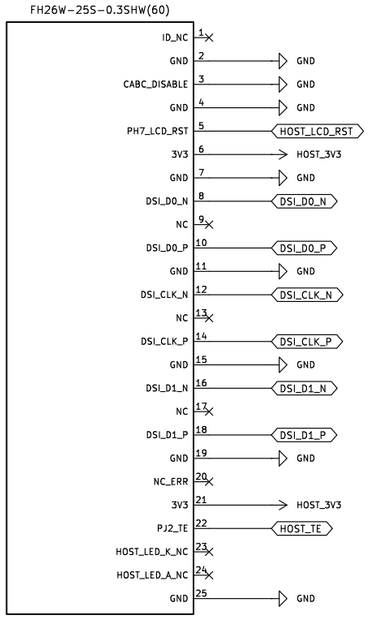
01 / STM32F469I-DISCO > DISEN LCD A / LCD B

Shared 39-pin mapping | A: DS070HSD39T-132 | B: DS0700B0E1A39CO-0180A (electrical approval pending)

01 / STM32 HOST - CN10



25-pin FPC / 0.30 mm pitch

HOST CONNECTION NOTES

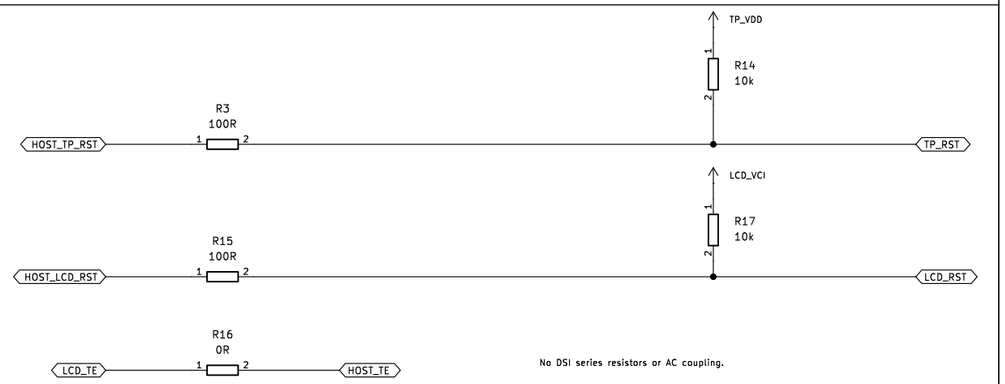
Remove original LCD and touch flexes.
Use purpose-made 1:1 FPC cables.

J1.23 / J1.24: host LED outputs unused.
J1.3 = GND disables default CABC control.
UM1932: R117 fitted / R119 absent.
Check actual host revision and population.

02 / MIPI DSI - DIRECT PAIRS

NET / PAIR	HOST J1	LCD J2
DSI_D0_P	10	16
DSI_D0_N	8	17
DSI_D1_P	18	20
DSI_D1_N	16	21
DSI_CLK_P	14	24
DSI_CLK_N	12	25

03 / RESET & FRAME SYNCHRONIZATION



No DSI series resistors or AC coupling.

05 / POWER, TOUCH & BACKLIGHT SHEETS

A: 8S2P / 21.6-27.2 V, 40 mA / ILI2130

B: 4S2P / 12 V typ, 40 mA / GT911

Logic power & touch

File: Logic_Touch.kicad_sch

Backlight / external power

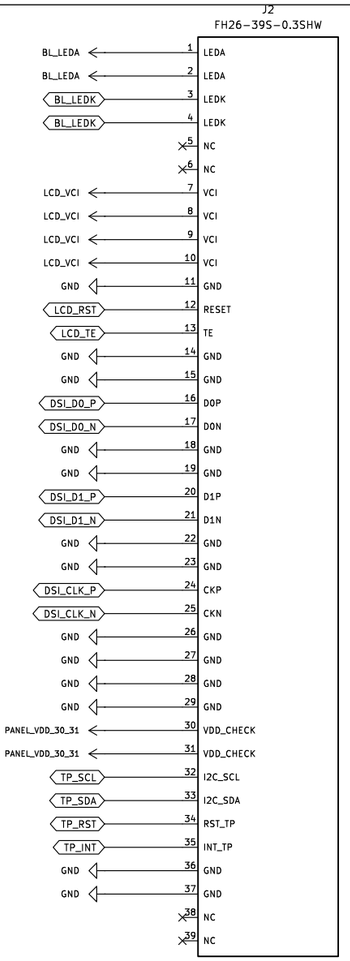
File: Backlight.kicad_sch

LCD configuration / HOLD

File: Panel_Selection.kicad_sch

One panel only. ALL supplies OFF before changing panel or JP1.
See sheet 4 for assembly links and conditional B approval.

04 / DISEN LCD - 39-PIN FPC



0.30 mm pitch / verify contact side & pin 1

Two data lanes + clock; direct pairs, no selector.
100 ohm differential target; actual stackup needed.
J2.30/31 isolated by R23 (factory DNP).
B VCI/VDD not specified: do not energize B.

REV K / Conditional dual-LCD prototype. Native configuration table is sheet 4.
Default HOLD: R4/R23 DNP, JP1 shunt absent. This does not isolate every panel signal or LED supply.

One LCD at a time | R4/R23 factory HOLD | See configuration sheet

Yakup Turkan

Sheet: /
File: STM32F469_DISEN_RevK.kicad_sch

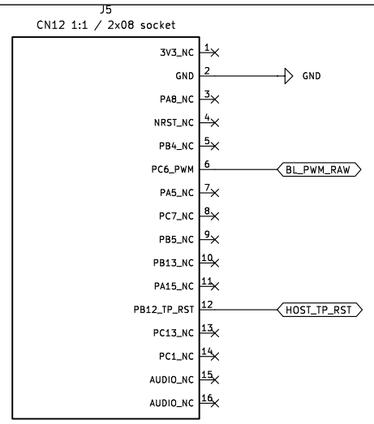
Title: STM32F469 / DISEN - MIPI & LCD interface

Size: A3	Date: 2026-10-19	**Rev: K / Dual LCD**
KiCad E.D.A. 9.0.2		Id: 1/4

02 / LOGIC POWER, TOUCH & HOST CONTROL

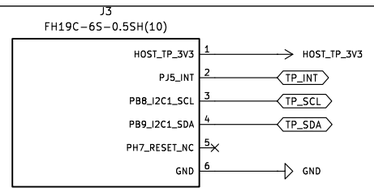
3.3 V logic domains | LCD and touch reset kept independent | Test points grouped by function

01 / STM32 CONTROL – CN12



2x8 socket / 2.54 mm / 1:1 mapping
Follow physical pin 1 for socket orientation.

02 / TOUCH HOST – CN2



HOST CN2 INPUT ONLY / 6-pin FPC
Never insert either panel touch tail into J3.
J3.1=3V3 / .6=GND; panel tail .1=GND / .6=3V3.

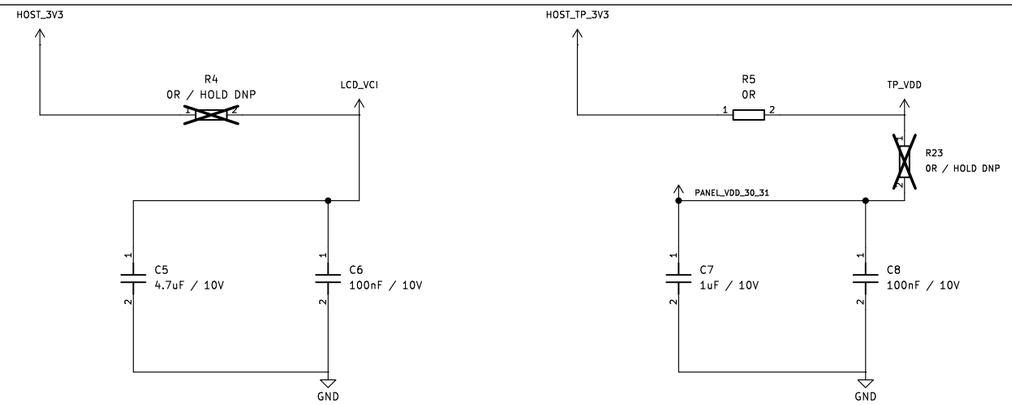
INTERFACE CONVENTIONS

Host LCD 3.3 V and host touch 3.3 V
remain distinct input nets.

BL_PWM_RAW: PC6 from CN12.
HOST_TP_RST: PB12 from CN12.
TP_INT: PJ5 from CN2.

Power symbols retain original rail names.

03 / LOGIC SUPPLIES & DECOUPLING

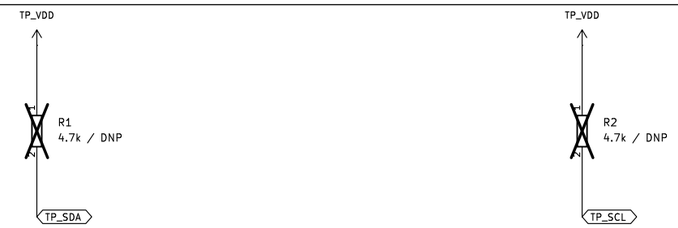


R4: VCI approval link (DNP)

R23: Isolated J2.30/31 supply link

Factory HOLD: R4/R23 DNP. Fit links only per sheet 4: unknown rails must remain separate.

04 / TOUCH I2C – OPTIONAL PULL-UPS

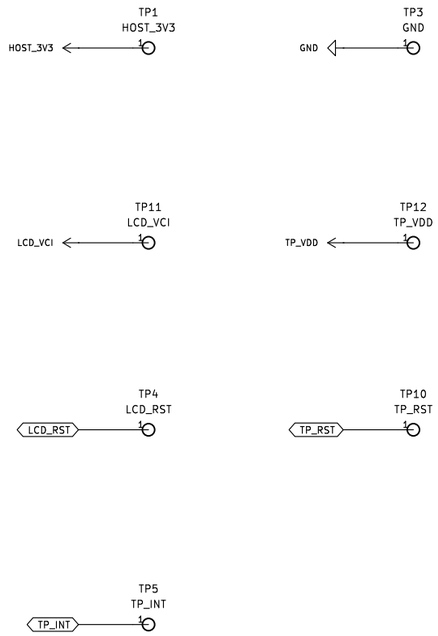


PB8/PB9 = I2C1; PJ5 = INT. R1/R2 remain optional DNP.
A uses ILI2130; B uses GT911. INT may need host drive during GT911 startup.
Confirm controller timing/address and host pull-ups; no firmware included.

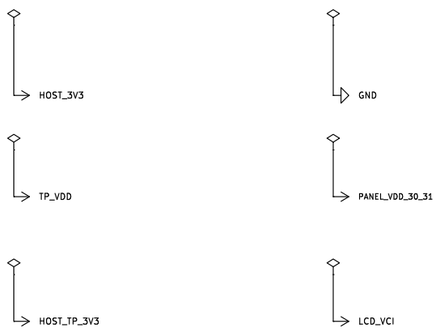
VALIDATION NOTES – REVISION K

A: VCI 3.0–3.6 V, touch 3.3 V nominal.
B: VCI and J2.30/31 VDD ranges UNKNOWN.
Both touch drawings: 2.8–3.3 V, supply current not given.
Resolve rail budgets, tolerance and reset timing before power-up.

05 / LOGIC TEST POINTS



Supply declarations



One LCD at a time | R4/R23 factory HOLD | See configuration sheet

Yakup Turkan

Sheet: /Logic power & touch/
File: Logic_Touch.kicad_sch

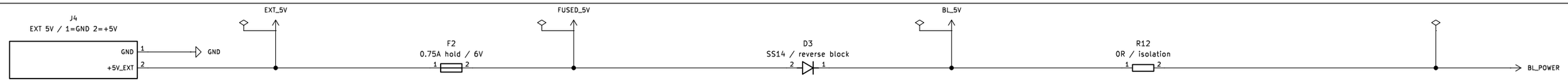
Title: STM32F469 / DISEN – Logic power & touch

Size: A3	Date: 2026-10-19	**Rev: K / Dual LCD**
KiCad E.D.A. 9.0.2		Id: 2/4

03 / EXTERNAL 5 V & TPS61165 BACKLIGHT

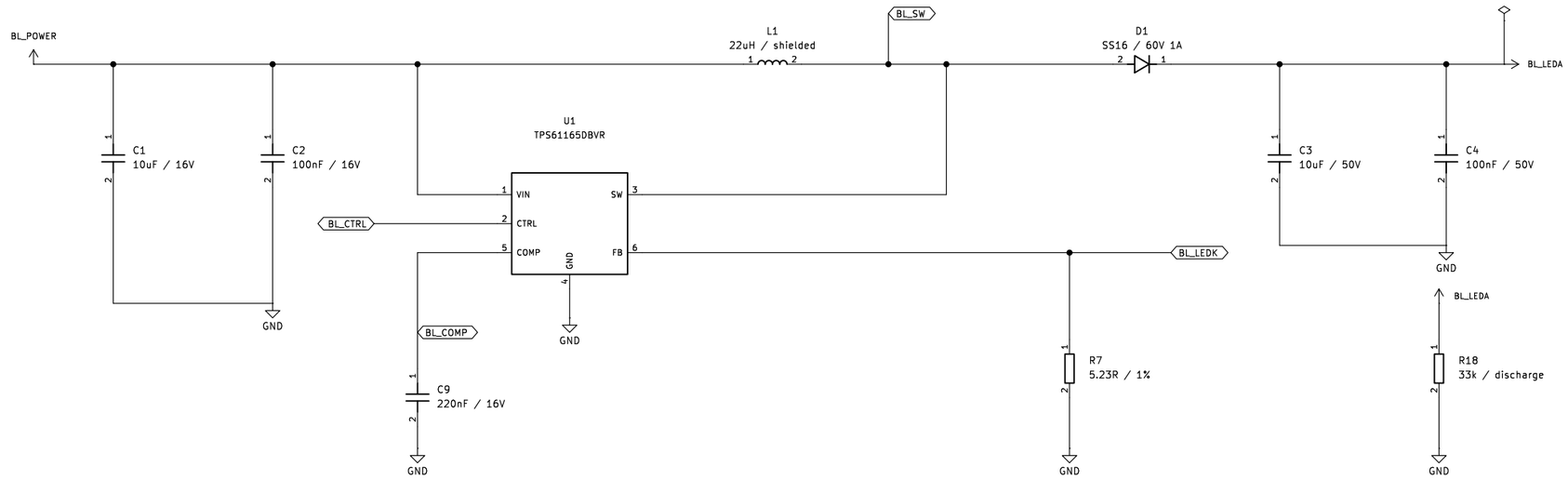
A: 21.6–27.2 V / 40 mA | B: 12 V typ / 40 mA | Common 38.24 mA setting; only OVP threshold is selected

01 / INPUT POWER – FUSE, REVERSE BLOCK & ISOLATION



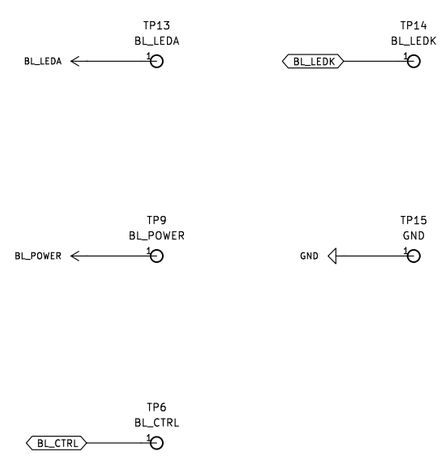
External regulated 5 V is mandatory. J4.1=GND / J4.2=+5V. No connection to host +5 V or host LEDA/LEDK.

02 / BOOST POWER STAGE & LED CURRENT RETURN

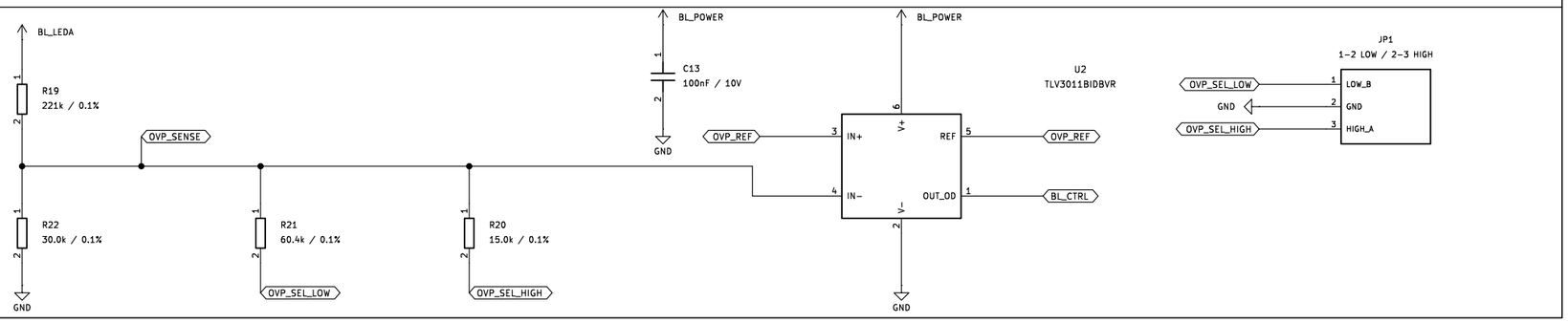


I_LED = 0.200 / 5.23 = 38.24 mA nominal; worst-case full-scale = 39.40 mA (TI VFB + R7 tolerance).

03 / BACKLIGHT TEST POINTS

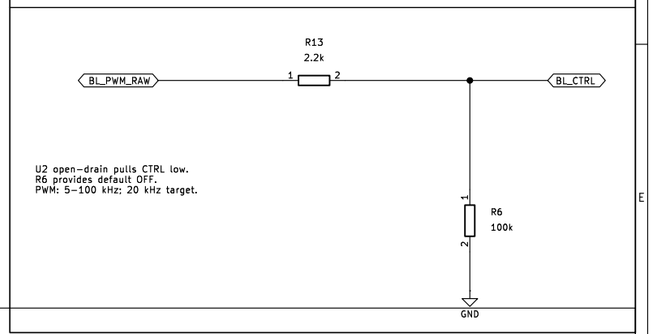


04 / SELECTABLE OVP – NOT AN OUTPUT VOLTAGE SETTING



JP1 is OVP selection only. R7 stays 5.23 ohm for BOTH panels: nominal 38.24 mA total. No high-current mode.
No shunt: 10.39 V nominal fallback; NOT a hard OFF. CTRL shutdown/filter delay and open-load peaks need measurement.
B LOW = 14.94 V provisional; B maximum Vf and absolute ratings are not supplied. See sheet 4 before fitting power links.

05 / PWM & DEFAULT-OFF



One LCD at a time | R4/R23 factory HOLD | See configuration sheet

Yakup Turkan

Sheet: /Backlight / external power/
File: Backlight.kicad_sch

Title: STM32F469 / DISEN – External 5 V & backlight

Size: A3	Date: 2026-10-19	**Rev: K / Dual LCD**
KiCad E.D.A. 9.0.2		Id: 3/4

04 / DUAL LCD CONFIGURATION & RELEASE CONDITIONS

One LCD at a time. Change cables/shunt only with ALL supplies OFF. Default assembly: R4/R23 DNP, no JP1 shunt.

ITEM	LCD A / HSD	LCD B / BOE – CONDITIONAL
Model	DS070HSD39T–132 / A0 preliminary	DS0700BOE1A39CO–0180A / drawing A01
Display / IC	280 x 1424 / OTA7290B	280 x 1424 / JD9365DA–H3
Touch IC	ILI2130 / I2C	GT911 / I2C; startup INT behavior needs review
39–pin / MIPI	D0 +/– 16/17, D1 +/– 20/21, CLK +/– 24/25	Same numbered lane/polarity assignments
LED array / current	8S2P / 21.6–27.2 V / IF 40 mA	4S2P / VF 12 V typ / IF 40 mA stated
VCI / J2.7–10	3.0–3.6 V documented	NO RANGE GIVEN – manufacturer confirmation
VDD / J2.30–31	3.3 V touch supply per A electrical table	VDD only; purpose/range NOT confirmed
Touch 6–pin tail	1 GND, 2 SCL, 3 SDA, 4 RST, 5 INT, 6 3V3	Same tail pin order; NOT compatible with host J3
Lens / thickness	48.25 x 216.54 / 5.39 mm nominal	56.95 x 216.46 / 5.20 mm nominal
Evidence limit	Currents TBD; physical flex mating unverified	Only mechanical drawing; Vf max / ratings missing

SETTING	LCD A	LCD B – DO NOT ENERGIZE UNTIL APPROVED
JP1 / nominal OVP	2–3 HIGH / 28.690 V	1–2 LOW / 14.936 V PROVISIONAL
R7 current sense	5.23 ohm 1% FIT / 38.24 mA nominal	Same R7; no doubling for parallel strings
R4 VCI link	Fit 0R after host power–budget review	DNP until 3.3 V VCI compatibility confirmed
R23 VDD link	Fit 0R for documented A 3.3 V touch rail	DNP until pin 30/31 voltage/purpose confirmed

RELEASE CONDITIONS / FACTORY HOLD

R4/R23 DNP is an assembly hold, NOT full isolation; I/O and LEDA can still be connected.
Do not connect B until VCI, VDD, LED Vf max and total IF are confirmed.
One shunt only. Missing shunt: ~10.39 V OVP fallback, not a shutdown interlock.
JP1 selects protection threshold; U1 still regulates CURRENT. OVP transients need tests.
J3 is HOST CN2 ONLY: .1=3V3 / .6=GND; panel touch tail .1=GND / .6=3V3.
Never plug a panel tail into J3. Use J2.32–35 only after internal routing is confirmed.
Review GT911 startup INT behavior, both supply budgets and exact TI U2 limits.

BEFORE POWER–UP

Panel B supply / LED ratings remain OPEN.
Default links DNP; see release conditions.

Documented settings; panel B electrical approval remains OPEN

Yakup Turkan

Sheet: /LCD configuration / HOLD/
File: Panel_Selection.kicad_sch

Title: STM32F469 / DISEN – Dual LCD configuration

Size: A3	Date: 2026–10–19	**Rev: K / HOLD**
KiCad E.D.A. 9.0.2		Id: 4/4